No.	Status	Date
LNDE-RPT-01-037-DS-2-RV-102	Issued	2001-03-02

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 2-RV-102 located within the ATU-100. The scope of work is limited to the thickness survey of this specific regenerator overhead line to assess wall loss and determine remaining life.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and recorded the data. The technician is responsible for the accurate placement of transducers and the recording of actual wall thicknesses.

Technical Basis and Procedures

The thickness measurements were performed under QSCP-NDT-001 Rev 0, the site NDE procedure issued for ATU-100 pre-startup and in-service NDE. All ultrasonic testing was conducted in accordance with the requirements of ASTM E797.

The data collected for this line is integrated into the broader 5-year Class 1 survey for sour gas, regenerator overhead and rich amine piping as detailed in LNDE-RPT-01-037.

Component Design Data

The subject line, 2-RV-102, is constructed from A106 Gr.B with an outer diameter of 60.3 mm and a nominal wall thickness of 5.54 mm. The minimum allowable thickness (t-min) is 2.79 mm.

Design parameters for this section of the ATU-100 are established at a maximum allowable pressure of 345 kPa(g) and a maximum allowable temperature of 121 °C.

Evaluation Criteria

Acceptance is based on the retirement thickness, which is defined as the t-min of 2.79 mm. Any reading recorded below this value shall be raised as an NCR.

Corrosion rates are calculated using the larger of the long-term rate since the first reading or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is determined by dividing the difference between the current thickness and t-min by the calculated rate.

Inspection Interval

As a Class 1 line (sour gas, regenerator overhead), the maximum interval for thickness and external visual inspections is 5 years.

The next thickness reading is due at no more than half the calculated remaining life, provided it does not exceed the 5 year class maximum.

Thickness readings, measured 2001-03-02

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	1996-03-08	7.12	6.82	0.060	0.060	>50	2006-03-02	
6-SG-101-02	7.11	3.56	1996-03-08	7.05	6.71	0.068	0.068	46.2	2006-03-02	
6-SG-101-03	7.11	3.56	1996-03-08	7.08	6.74	0.068	0.068	46.6	2006-03-02	
6-SG-101-04	7.11	3.56	1996-03-08	7.09	6.73	0.072	0.072	43.9	2006-03-02	
6-SG-101-05	7.11	3.56	1996-03-08	7.14	6.68	0.092	0.092	33.8	2006-03-02	
6-SG-101-06	7.11	3.56	1996-03-08	7.07	6.87	0.040	0.040	>50	2006-03-02	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	1996-03-08	7.14	6.72	0.084	0.084	37.5	2006-03-02	
6-SG-102-02	7.11	3.56	1996-03-08	7.11	6.72	0.078	0.078	40.4	2006-03-02	
6-SG-102-03	7.11	3.56	1996-03-08	7.04	6.69	0.070	0.070	44.6	2006-03-02	
6-SG-102-04	7.11	3.56	1996-03-08	6.99	6.64	0.070	0.070	43.8	2006-03-02	
6-SG-102-05	7.11	3.56	1996-03-08	7.13	6.93	0.040	0.040	>50	2006-03-02	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	1996-03-08	8.52	8.39	0.026	0.026	>50	2011-03-02	
4-RA-101-02	8.56	3.81	1996-03-08	8.54	8.45	0.018	0.018	>50	2011-03-02	
4-RA-101-03	8.56	3.81	1996-03-08	8.61	8.42	0.038	0.038	>50	2011-03-02	
4-RA-101-04	8.56	3.81	1996-03-08	8.63	8.37	0.052	0.052	>50	2011-03-02	
4-RA-101-05	8.56	3.81	1996-03-08	8.60	8.42	0.036	0.036	>50	2011-03-02	
4-RA-101-07	8.56	3.81	1996-03-08	8.60	8.51	0.018	0.018	>50	2011-03-02	
4-RA-101-08	8.56	3.81	1996-03-08	8.68	8.30	0.076	0.076	>50	2011-03-02	
4-RA-101-09	8.56	3.81	1996-03-08	8.57	8.43	0.028	0.028	>50	2011-03-02	
4-RA-101-10	8.56	3.81	1996-03-08	8.67	8.48	0.038	0.038	>50	2011-03-02	
4-RA-101-11	8.56	3.81	1996-03-08	8.68	8.39	0.058	0.058	>50	2011-03-02	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	1996-03-08	8.54	8.48	0.012	0.012	>50	2011-03-02	
4-RA-102-02	8.56	3.81	1996-03-08	8.57	8.50	0.014	0.014	>50	2011-03-02	
4-RA-102-03	8.56	3.81	1996-03-08	8.56	8.37	0.038	0.038	>50	2011-03-02	
4-RA-102-04	8.56	3.81	1996-03-08	8.54	8.35	0.038	0.038	>50	2011-03-02	
4-RA-102-05	8.56	3.81	1996-03-08	8.55	8.36	0.038	0.038	>50	2011-03-02	
4-RA-102-06	8.56	3.81	1996-03-08	8.51	8.38	0.026	0.026	>50	2011-03-02	
4-RA-102-07	8.56	3.81	1996-03-08	8.58	8.49	0.018	0.018	>50	2011-03-02	
4-RA-102-08	8.56	3.81	1996-03-08	8.49	8.36	0.026	0.026	>50	2011-03-02	
4-RA-102-09	8.56	3.81	1996-03-08	8.55	8.35	0.040	0.040	>50	2011-03-02	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	1996-03-08	11.00	10.82	0.036	0.036	>50	2011-03-02	
6-RA-103-02	10.97	4.57	1996-03-08	11.09	10.69	0.080	0.080	>50	2011-03-02	
6-RA-103-03	10.97	4.57	1996-03-08	10.99	10.84	0.030	0.030	>50	2011-03-02	
6-RA-103-04	10.97	4.57	1996-03-08	10.98	10.59	0.078	0.078	>50	2011-03-02	
6-RA-103-05	10.97	4.57	1996-03-08	10.95	10.61	0.068	0.068	>50	2011-03-02	
6-RA-103-06	10.97	4.57	1996-03-08	11.01	10.86	0.030	0.030	>50	2011-03-02	
6-RA-103-07	10.97	4.57	1996-03-08	10.99	10.79	0.040	0.040	>50	2011-03-02	
6-RA-103-08	10.97	4.57	1996-03-08	10.97	10.83	0.028	0.028	>50	2011-03-02	
6-RA-103-09	10.97	4.57	1996-03-08	11.01	10.85	0.032	0.032	>50	2011-03-02	
6-RA-103-10	10.97	4.57	1996-03-08	11.02	10.78	0.048	0.048	>50	2011-03-02	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	1996-03-08	7.59	7.26	0.066	0.066	>50	2006-03-02	
3-RV-101-02	7.62	3.56	1996-03-08	7.57	7.31	0.052	0.052	>50	2006-03-02	
3-RV-101-03	7.62	3.56	1996-03-08	7.52	7.15	0.074	0.074	48.3	2006-03-02	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-04	7.62	3.56	1996-03-08	7.59	7.18	0.082	0.082	44.0	2006-03-02	
3-RV-101-05	7.62	3.56	1996-03-08	7.67	7.25	0.084	0.084	43.8	2006-03-02	
3-RV-101-06	7.62	3.56	1996-03-08	7.59	7.20	0.078	0.078	46.5	2006-03-02	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	1996-03-08	5.53	5.29	0.048	0.048	>50	2006-03-02	
2-RV-102-02	5.54	2.79	1996-03-08	5.59	5.12	0.094	0.094	24.7	2006-03-02	
2-RV-102-03	5.54	2.79	1996-03-08	5.48	5.24	0.048	0.048	>50	2006-03-02	
2-RV-102-04	5.54	2.79	1996-03-08	5.58	5.33	0.050	0.050	>50	2006-03-02	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	1996-03-08	8.51	8.40	0.022	0.022	>50	2011-03-02	
4-RA-104-02	8.56	3.81	1996-03-08	8.55	8.41	0.028	0.028	>50	2011-03-02	
4-RA-104-03	8.56	3.81	1996-03-08	8.54	8.46	0.016	0.016	>50	2011-03-02	
4-RA-104-04	8.56	3.81	1996-03-08	8.59	8.35	0.048	0.048	>50	2011-03-02	
4-RA-104-05	8.56	3.81	1996-03-08	8.60	8.29	0.062	0.062	>50	2011-03-02	
4-RA-104-06	8.56	3.81	1996-03-08	8.59	8.39	0.040	0.040	>50	2011-03-02	
4-RA-104-07	8.56	3.81	1996-03-08	8.59	8.29	0.060	0.060	>50	2011-03-02	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	2001-03-02	_____